

Application of Built-in Adaptive Remeshing and Mesh Refinement Features in Abaqus to Fracture Simulations Using Phase-field User Elements

Supervisor-Aligned Integrated Thesis Proposal and Execution Plan

Working revision integrating the formal proposal of 30 June 2026 with supervisor feedback of 3
September 2026

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Document status	Execution addendum - does not replace the signed original proposal

**Governing rule: Understand the complete Mode-I benchmark before increasing
complexity.**

**Define problem -> define expected solution -> establish reference -> apply adaptive method
-> compare -> explain discrepancies**

1. Purpose, authority, and source hierarchy

Status and authority. This document preserves all ten tasks of the formal Master's Thesis Proposal. The supervisor meeting adds a sequencing and evidence-quality gate: the Pandey-Kumar Mode-I benchmark must be understood, reproduced, and explained before Mode-II, mixed-mode, or other higher-complexity branches resume. It is an execution addendum, not a replacement of the signed proposal.

The integrated execution plan uses the following hierarchy whenever two documents appear to pull the work in different directions:

Priority	Role
1 - Current supervisor instruction	Controls immediate sequencing, scientific questions, and what must be understood before complexity is increased.
2 - Formal thesis proposal (30 June 2026)	Defines the mandatory end goals. Tasks 4, 5, and 6 cannot be replaced by adjacent technical work.
3 - Primary literature	Defines the reference theory, benchmark configuration, Abaqus implementation ideas, and reference workflow.
4 - Verified project evidence	Defines what the current code, meshes, solver jobs, and extracted fields have actually demonstrated.
5 - Secondary summaries / AI assistance	May organize information but may not override primary sources or verified evidence.

2. Original proposal preserved: mandatory tasks

The original proposal remains the governing scope. The supervisor meeting does not delete or replace any task; it changes the order in which complexity is allowed to increase and raises the evidence standard for closing the Mode-I work.

Task	Formal scope	Supervisor-aligned execution status
1	Familiarization with the topic "Phase-field for Fracture"	Foundation
2	Literature review	Foundation
3	Reproduction of the simulations performed in the reference paper without mesh refinement using the available user code [Molnar & Gravouil]	Mode-I gate
4	Python implementation of	Mode-I gate

Task	Formal scope	Supervisor-aligned execution status
	mesh refinement and adaptive remeshing as described in the reference paper [Pandey & Kumar]	
5	Reproduction of the results from the reference paper with mesh refinement	Mode-I gate
6	Integration of the ABAQUSER visualization tool developed at IMFD into the modeling framework	Mandatory after Mode-I closure
7	Application to benchmark problems in fracture mechanics; conducting sensitivity studies	Only after Mode-I closure
8	Formulation of recommendations for selecting the parameters of the meshing tool	After evidence synthesis
9	Documentation for future users	After workflow is stable
10	Writing the Master's Thesis	Continuous, then final synthesis

3. Literature integration map

The primary literature has distinct roles. Keeping these roles separate prevents the thesis from attributing a method or tool to the wrong paper.

Source	What it legitimately supports	Proposal role
Msekh et al. (2015)	Abaqus implementation of a phase-field model for brittle fracture using UEL and UMAT. Important theoretical/implementation foundation and overlay/post-processing concept.	Tasks 1-3
Molnar & Gravouil (2017)	Robust staggered phase-field solution in Abaqus/Standard using UEL; supplied source/tutorials form the practical baseline code used for	Tasks 1-3

Source	What it legitimately supports	Proposal role
	verification.	
Pandey & Kumar (2025)	Core adaptive-refinement reference: MISESRI-based error indication, RemeshingRule, adaptiveRemesh, Mode-I benchmark, and automated Python workflow. Listing 1 uses errorTarget=1.0 and refinementFactor=10; Mode-I proposed mesh reported as 13,941 elements.	Tasks 4-5, later Task 8
Diddige, Roth & Kiefer (2025)	IMFD phase-field implementation context. The paper explicitly reports use of an in-house ABAQUSER visualization tool for post-processing and cites the dedicated ABAQUSER/Viewer work.	Task 6 context
Adaptive Remeshing PFF Rapid Study Guide	Secondary project study aid only. Useful for orientation, but primary papers and verified model evidence control scientific claims.	Support only

Important Task-6 boundary: Diddige et al. supports the existence/use of the IMFD ABAQUSER visualization path, but it should not be used to invent the internal implementation of the tool. The authentic institute tool must be obtained or its interface verified from institute documentation/software.

4. Active scientific focus: the Pandey-Kumar Mode-I benchmark

The supervisor asked that the simplest benchmark be understood completely before complexity is increased. The active boundary-value problem is therefore fixed as follows:

Geometry	1 mm x 1 mm square plate.
Initial crack	0.5 mm sharp edge crack / zero-gap seam. Do not use a finite-width slot.
Loading	Mode-I, displacement controlled.
Elasticity	$E = 210 \text{ GPa}$, $\nu = 0.3$.
Phase-field parameters	$l_0 = 0.0075 \text{ mm}$; $G_c = 2.7 \times 10^{-3} \text{ kN/mm}$.
Expected crack behaviour	Crack-tip localization followed by horizontal

	extension across the ligament.
Reference response anchor	Digitized literature target about $F_{\max} = 0.758$ kN at u approximately 0.005860 mm; reconstructed fixed-mesh response 0.757778 kN at 0.005857 mm.

Expected response before looking at the adaptive result

- Initial approximately linear global force-displacement response.
- Localization of the phase-field near the crack tip as the tensile driving energy increases.
- Peak reaction force at fracture initiation/localization, followed by post-peak softening and strong load reduction.
- Damage/crack path extending approximately horizontally for this symmetric Mode-I configuration.
- Refinement concentrated where the coarse stress field is insufficiently resolved, rather than uniformly over the entire domain.

Anchor statement: This fixed-mesh response is the reference that every adaptive result must reproduce to an acceptable, predeclared accuracy before efficiency is discussed.

5. Evidence standard: convergence is multi-quantity, not peak-force only

The meeting explicitly rejected peak reaction force as a sufficient convergence criterion. The Mode-I study must assess global, local, and computational evidence together.

Required quantity	How it is used
Complete F-u curve	Compare elastic slope, approach to peak, peak, softening, and final load level on a common displacement axis. Use a curve error metric such as NRMSE or normalized L2 in addition to plots.
Peak force $F_{\max}$	Report value and relative difference to both the fixed-mesh anchor and digitized paper target.
Displacement at peak	Report $u(F_{\max})$; this captures whether localization is delayed or premature.
Initial structural stiffness K_0	Define as the slope dF/du in the initial elastic range. K_0 is global specimen stiffness, not Young's modulus.
Phase-field / damage field	Compare d contours at matched displacements and, where practical, sampled d -profiles through the process zone.
Crack path / localization profile	Compare path geometry separately from global force agreement; one does not guarantee the other.

Required quantity	How it is used
Energy / work	Energy convergence is desirable. The current UEL does not provide a qualified complete Abaqus internal-energy balance. If external work integral $W_{ext} = \int F \, du$ is used as a global response metric, label it explicitly and do not substitute it for missing UEL internal/fracture energy.
Computational cost	Element count, increments, iterations, cutbacks, CPU time, walltime, peak memory. Efficiency claims are allowed only after scientific equivalence is demonstrated.

Acceptance thresholds must be written before examining a new production result. Do not retrofit thresholds after seeing the answer. Literature-digitization uncertainty and supervisor agreement should be recorded when defining the final gates.

6. MISESeri and Abaqus remeshing: the mechanism that must be understood

Pandey and Kumar use Abaqus recovered-stress error estimation as the driver for pre-refinement. In the current workflow, the companion standard/UMAT representation exposes the stress field so Abaqus can evaluate MISESeri.

Coarse finite-element stress field
Stress recovery / discretization-error estimation
MISESeri element-field indicator
RemeshingRule with declared error target and size controls
Abaqus sizing decision
adaptiveRemesh generates a changed mesh
Reconstruct co-located UEL and companion visualization layers, then run the fracture calculation

MISESeri is a stress-discretization error indicator. It is not a phase-field error and not a damage error. In the project Figure 3.1, the plotted values come from an element-field extraction of the coarse pre-analysis; they should not simply be described as "nodes".

Mandatory question before accepting any refined mesh: Does the spatial refinement pattern correspond to the locations and magnitudes suggested by the MISESeri field, and can the link from indicator to sizing decision be explained from Abaqus settings and source data?

7. Central technical problem: the nominal 1% mesh-density discrepancy

The current reconstruction and the paper do not produce the same mesh density under the nominal Mode-I settings. This is not to be hidden by selecting whichever errorTarget happens to give a similar element count.

Case	Element count	Scientific interpretation
Pandey-Kumar Mode-I proposed mesh	13,941	Published reference result.
Current nominal 1% reconstruction	71,320	Major unresolved discrepancy in sizing/refinement stage.
Current 2% sensitivity	15,396	Numerically close in element count, but not evidence that the paper used 2%.
Current 5% sensitivity	4,194	Coarser sensitivity only.

Root-cause audit questions - answer in this order

- Is the coarse geometry identical to the reference benchmark, especially the zero-gap seam and boundary conditions?
- Is the coarse mesh topology and seeding sufficiently faithful to the paper?
- Is the pre-analysis loading history and increment schedule identical to the intended reference workflow?
- Is MISESeri requested from the correct standard/facsimile element set and extracted at the correct location/step?
- Is the RemeshingRule applied to the same region/set and with the same sizing method, errorTarget, refinementFactor, minimum size, maximum size, and coarsening state?
- Is adaptiveRemesh followed by any operation that can overwrite or re-seed the resulting mesh?
- Does Abaqus version/release-dependent behaviour plausibly affect sizing? If tested, isolate version as the only changed variable.
- Can the paper itself fully specify all preprocessing details? If not, record the remaining uncertainty as unresolved rather than inventing a cause.

8. Supervisor-aligned gated execution plan

No new branch is allowed to bypass the active gate. A stage is closed only when its evidence packet is complete and the conclusion can be defended without relying on AI-generated interpretation.

Gate	Proposal mapping	Exit evidence	Hard stop rule
Gate 0 - Source and scope freeze	Tasks 1-2	Primary papers, formal proposal, meeting notes, and benchmark parameters are indexed. Active question is stated in one sentence.	Do not run simulations until the exact Mode-I problem and expected response are written down.

Gate	Proposal mapping	Exit evidence	Hard stop rule
Gate 1 - Conventional Mode-I reference	Task 3	Fixed-mesh model reproduces the paper/reference response and its mesh/convergence basis is documented.	No adaptive interpretation before the reference solution is qualified.
Gate 2 - Multi-quantity convergence qualification	Task 3 prerequisite	F-u curve, Fmax, upeak, K0, d-field/profile, crack path, energy/work status, and computational cost are all addressed.	Peak force alone cannot close the gate.
Gate 3 - MISESeri mechanism verification	Task 4 prerequisite	Source of MISESeri, field meaning, element set, step, output position, and spatial distribution are understood and reproducible.	Do not treat MISESeri as phase-field error.
Gate 4 - Native Python refinement implementation	Task 4	RemeshingRule + adaptiveRemesh generate a genuine topology/size change and UEL/UMAT layers are reconstructed deterministically.	No custom remesher may substitute for the built-in reference method.
Gate 5 - Nominal 1% discrepancy audit	Tasks 4-5 prerequisite	71,320 vs 13,941 is either explained by a verified isolated cause or documented as unresolved due to missing reference information.	Do not tune to 2% merely to match count.
Gate 6 - Reference reproduction with refinement	Task 5	Refined run is compared to fixed reference and paper using the full evidence set. Scientific equivalence and efficiency are separated.	No efficiency claim without response equivalence.
Gate 7 - ABAQUSER integration	Task 6	Authentic IMFD ABAQUSER path is	Companion bridge work cannot silently

Gate	Proposal mapping	Exit evidence	Hard stop rule
		integrated and verified, or an explicit supervisor decision documents any accepted substitute/boundary.	replace the mandatory tool.
Gate 8 - Higher-complexity benchmarks	Task 7	Only after Gates 1-7. Resume Mode-II/mixed-mode/sensitivity studies with predeclared questions and isolated variables.	Mode-II remains on HOLD before this gate.
Gate 9 - Parameter recommendations	Task 8	Recommendations are based on verified sensitivity trends, not a single benchmark outcome.	State the applicability range and limitations.
Gate 10 - Future-user documentation	Task 9	Reproduction package contains .inp, .for, .py, commands.txt, environment notes, and evidence provenance.	Do not require multi-GB ODB transfer for basic reproduction.
Gate 11 - Thesis synthesis	Task 10	Final thesis reorganizes by scientific logic, not historical chronology; verified and unresolved findings are clearly separated.	No unresolved question is written as a proven mechanism.

9. Anti-deviation rules for every simulation, script change, and report update

- Every action must identify one original proposal task number and one active gate. If it cannot, do not perform it.
- Before a new solver job: write the hypothesis, the single intended variable change, expected effect, acceptance quantities, and exact inputs/hashes.
- Change one scientific/numerical factor at a time whenever the purpose is diagnosis. Do not change geometry, mesh rule, time stepping, material parameters, and solver controls simultaneously.
- Queue availability or spare HPC capacity is never by itself a scientific reason to launch a job.
- A job that exits successfully is technically complete, not automatically scientifically accepted.
- Unexpected results trigger the smallest evidence-supported diagnostic, not a new branch.
- Mode-II and multi-step topology-changing restart work remain archived until the Mode-I gate is closed.

- Do not relabel 2% as the literature setting. Pandey and Kumar explicitly report errorTarget=1.0 in the core listing and separately study other values.
- Use standard terminology: "finite elements" or "element count". Avoid "physical elements" except where a precise layer distinction is explicitly defined and justified.
- Every report claim must be categorized as: known from source/theory; verified numerically; or unresolved.

Do-not-deviate test: If the proposed next action does not reduce uncertainty in the active Mode-I gate, it is postponed unless the supervisor explicitly redirects the work.

10. Condensed Mode-I report requested by the supervisor

Chapter 1 - Benchmark problem - Geometry, material, loading, phase-field parameters, expected physical response, and reference literature. Start with the boundary-value problem, not with element-count discrepancies.

Chapter 2 - Conventional/reference solution - Fixed-mesh model, mesh basis, complete F-u response, convergence criteria, phase-field evolution, energy/work discussion, and computational cost. Establish the reference anchor.

Chapter 3 - Pandey-Kumar adaptive method - Explain the two-job chain: coarse model -> pre-analysis -> MISESeri -> RemeshingRule -> adaptiveRemesh -> refined mesh -> co-located layer reconstruction -> fracture solve.

Chapter 4 - Expected refinement behaviour and discrepancy analysis - State where refinement should occur before showing the result. Compare MISESeri spatial distribution with the actual refined mesh and perform the 1% root-cause audit.

Chapter 5 - Actual adaptive result and conclusions - Compare mesh, F-u, Fmax, upeak, K0, d-field, crack path, cost, and literature. Separate VERIFIED findings from UNRESOLVED questions. State exactly what can and cannot be concluded.

11. Supervisor reproduction package

The supervisor specifically requested lightweight inputs and commands rather than large ODB transfers. The package should be reproducible on the institute system.

[] Mode-I fixed-reference input deck (.inp)

[] Coarse pre-analysis input deck (.inp)

[] Refined production input deck (.inp)

[] Fortran user-subroutine source(s) (.for)

[] Abaqus/CAE Python remeshing script(s) (.py)

[] Post-processing/extraction script(s) (.py)

[] commands.txt with the exact real execution sequence and required module/compiler setup

[] README describing expected outputs and how to verify success

[] Small evidence tables/CSV needed for curves and MISESeri; avoid shipping multi-GB ODBs unless specifically requested

[] Version/environment record: Abaqus release, compiler, OS/HPC queue assumptions, serial restriction if applicable

12. Current working checkpoint and immediate next work

The September progress report already established a strong fixed-mesh Mode-I anchor and a working Abaqus-native pre-refinement chain, but it also documented the nominal 1% mesh-density discrepancy. The supervisor meeting therefore resets the active work from "more simulations" to "complete understanding of the Mode-I chain".

Already established / carry forward	Must be re-checked or completed now	Explicitly paused
Mode-I geometry and parameters; fixed-mesh response close to digitized reference; native adaptiveRemesh chain demonstrated; 1/2/5% element-count sensitivity exists.	Full reference convergence evidence; exact MISESeri-to-sizing understanding; root cause / limitation statement for 71,320 vs 13,941; refined response judged against all metrics; condensed Mode-I report and supervisor package.	Mode-II, mixed-mode extensions, new topology-changing restart campaigns, nonessential optimization, and any new branch not required to close the active Mode-I gate.

Immediate ordered work:

1. Freeze the Mode-I benchmark definition and produce a one-page expected-response sheet.
2. Rebuild the fixed-mesh convergence evidence using more than peak force: full F-u, upeak, K0, phase-field/profile, crack path, energy/work status, and cost.
3. Audit Figure 3.1 / MISESeri data: what each plotted point represents, output position, element set, step, and spatial pattern.
4. Trace Abaqus RemeshingRule settings and sizing path line-by-line against Pandey & Kumar; document every parameter and any unavoidable ambiguity.
5. Explain or formally bound the 71,320 vs 13,941 discrepancy using isolated tests only.
6. Only then judge the refined fracture result and move to Task 6.

13. Primary references used in this integrated plan

- Molnar, G.; Gravouil, A. (2017). 2D and 3D Abaqus implementation of a robust staggered phase-field solution for modeling brittle fracture. *Finite Elements in Analysis and Design*, 130, 27-38.
- Msekh, M. A.; Sargado, J. M.; Jamshidian, M.; Areias, P. M.; Rabczuk, T. (2015). Abaqus implementation of phase-field model for brittle fracture. *Computational Materials Science*, 96, 472-484.
- Pandey, A.; Kumar, S. (2025). A Simple and Robust Mesh Refinement Implementation in Abaqus for Phase Field Modelling of Brittle Fracture. *Computer Modeling in Engineering & Sciences*, 144(3), 3251-3286.
- Diddige, V.; Roth, S.; Kiefer, B. (2025). Phase-field modeling of hydrogen-promoted fracture: Natural incorporation of hydrostatic stress dependencies via a chemical potential-based variational formulation. *Computer Methods in Applied Mechanics and Engineering*, 445, 118143.
- Master's Thesis Proposal (30 June 2026), TU Bergakademie Freiberg, IMFD.
- Supervisor progress-report discussion meeting (3 September 2026) and the associated September progress report.

Appendix A - Master ordered checklist

This checklist is intentionally redundant with the separate printable checklist PDF. The active gate must be completed in order; later boxes remain untouched unless the supervisor explicitly redirects the sequence.

A. Scope freeze

Done	Requirement	Evidence / date
☐	Mode-I benchmark written in one page	
☐	Primary literature roles verified	
☐	Active open question written in one sentence	
☐	Mode-II and state-transfer extensions marked HOLD	

B. Fixed reference

Done	Requirement	Evidence / date
☐	Exact geometry/BC/materials verified	
☐	Reference mesh documented	
☐	Full F-u curve extracted	
☐	Fmax and upeak recorded	
☐	K0 method documented	
☐	Matched d contours/profiles prepared	
☐	Energy/work limitation documented	
☐	Computational cost recorded	

C. MISESRI understanding

Done	Requirement	Evidence / date
☐	MISESRI meaning stated correctly	
☐	Element-field output location	

Done	Requirement	Evidence / date
	verified	
☐	Correct set/step/output position verified	
☐	Spatial error pattern interpreted	
☐	Expected refinement region stated before viewing refined mesh	

D. Native remeshing

Done	Requirement	Evidence / date
☐	RemeshingRule parameters audited against paper	
☐	adaptiveRemesh proven to change mesh	
☐	No post-remesh generateMesh overwrite	
☐	UEL/UMAT layer reconstruction deterministic	
☐	Mesh sets/BCs/properties preserved	

E. 1% discrepancy

Done	Requirement	Evidence / date
☐	Coarse topology checked	
☐	Pre-analysis loading checked	
☐	Remeshing region checked	
☐	Version effect isolated if tested	
☐	1/2/5% sensitivity kept clearly separated	
☐	Cause verified OR uncertainty formally documented	

F. Refined reproduction

Done	Requirement	Evidence / date
☐	Full F-u compared	
☐	Fmax/upeak/K0 compared	
☐	Matched d contours/profiles compared	
☐	Crack path compared	
☐	Cost compared	
☐	Scientific equivalence concluded separately from efficiency	

G. Task 6

Done	Requirement	Evidence / date
☐	Authentic ABAQUSER tool/interface obtained	
☐	Integration procedure documented	
☐	Fields verified against independent extraction	
☐	If unavailable, supervisor decision on boundary recorded	

H. Later tasks

Done	Requirement	Evidence / date
☐	Mode-II resumed only after Mode-I closure	
☐	Sensitivity studies have predeclared questions	
☐	Meshing recommendations synthesized	
☐	Future-user package complete	
☐	Final thesis reorganized scientifically, not historically	

